# Weather System To Optimize Solar Car Routes.

Project Expo Presentation(PRJ381 Bcomp Group 5)

# 01 DEMO of FULL System

# 02 Technology stack diagram

## 1 Sensing hardware on lead vehicle

BME280 · Anemometer · GPS module · LED indicator

## 2 Edge device

Raspberry Pi · Python · JSON payload · MQTT + TLS

## 3 Cloud messaging & ingestion

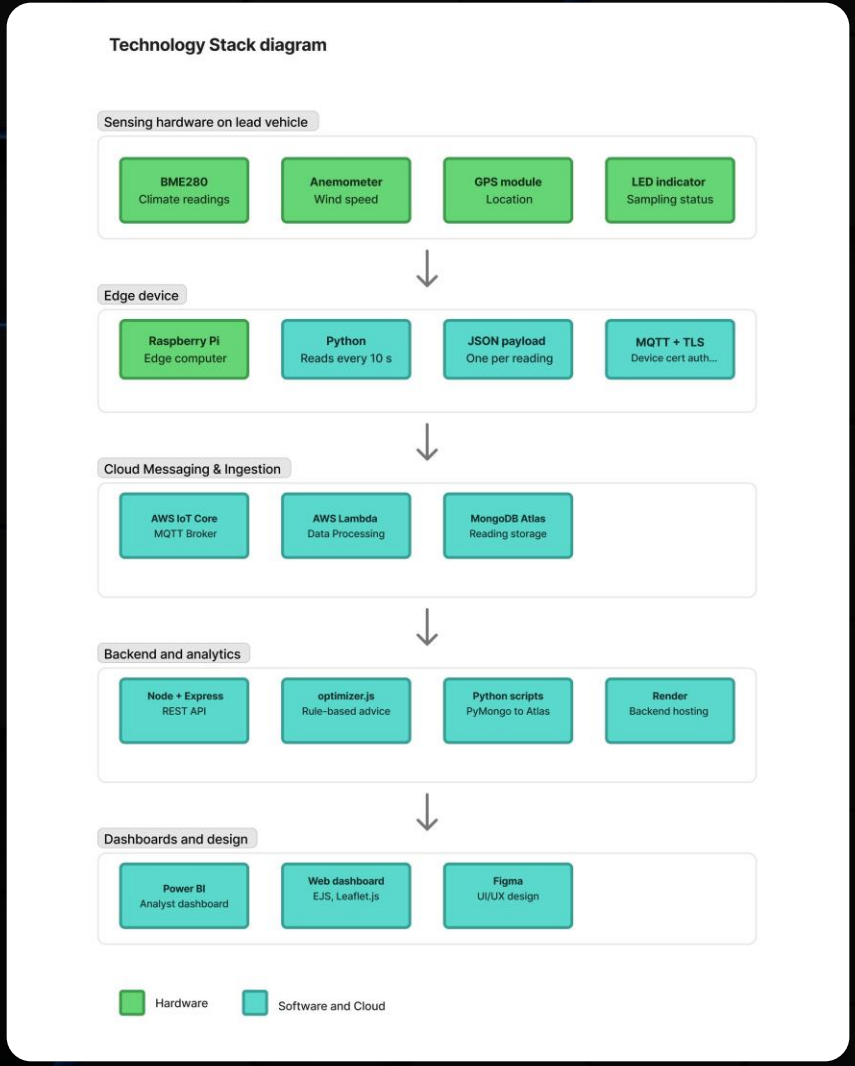
AWS IoT Core · AWS Lambda · MongoDB Atlas

## 4 Backend and analytics

Node + Express · optimizer.js · Python scripts · Render

## 5 Dashboards and design

Power BI · Web dashboard (EJS, Leaflet.js) · Figma



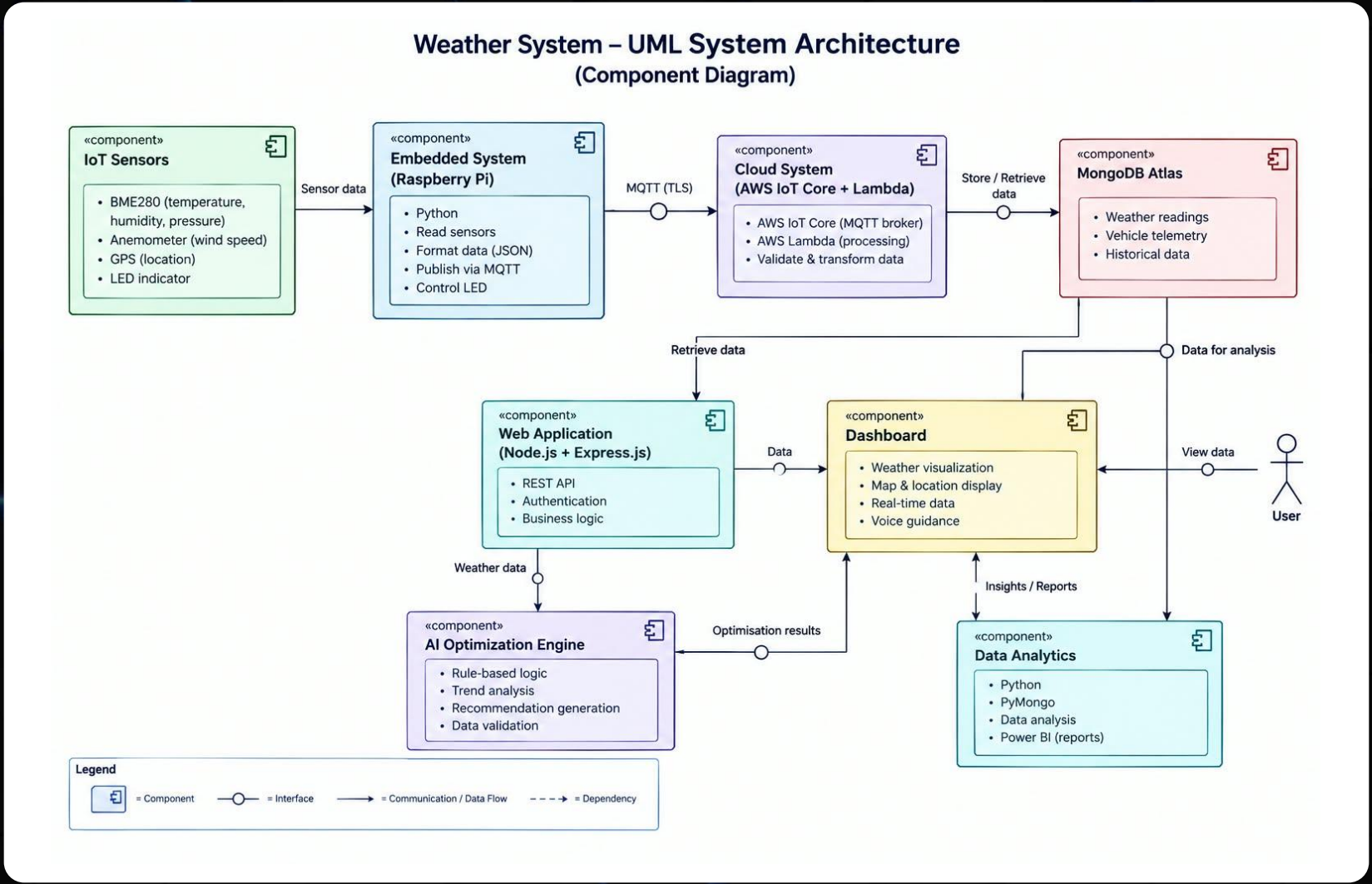
# 03 System Architecture

## UML diagram

of the overall system

## Subsystems

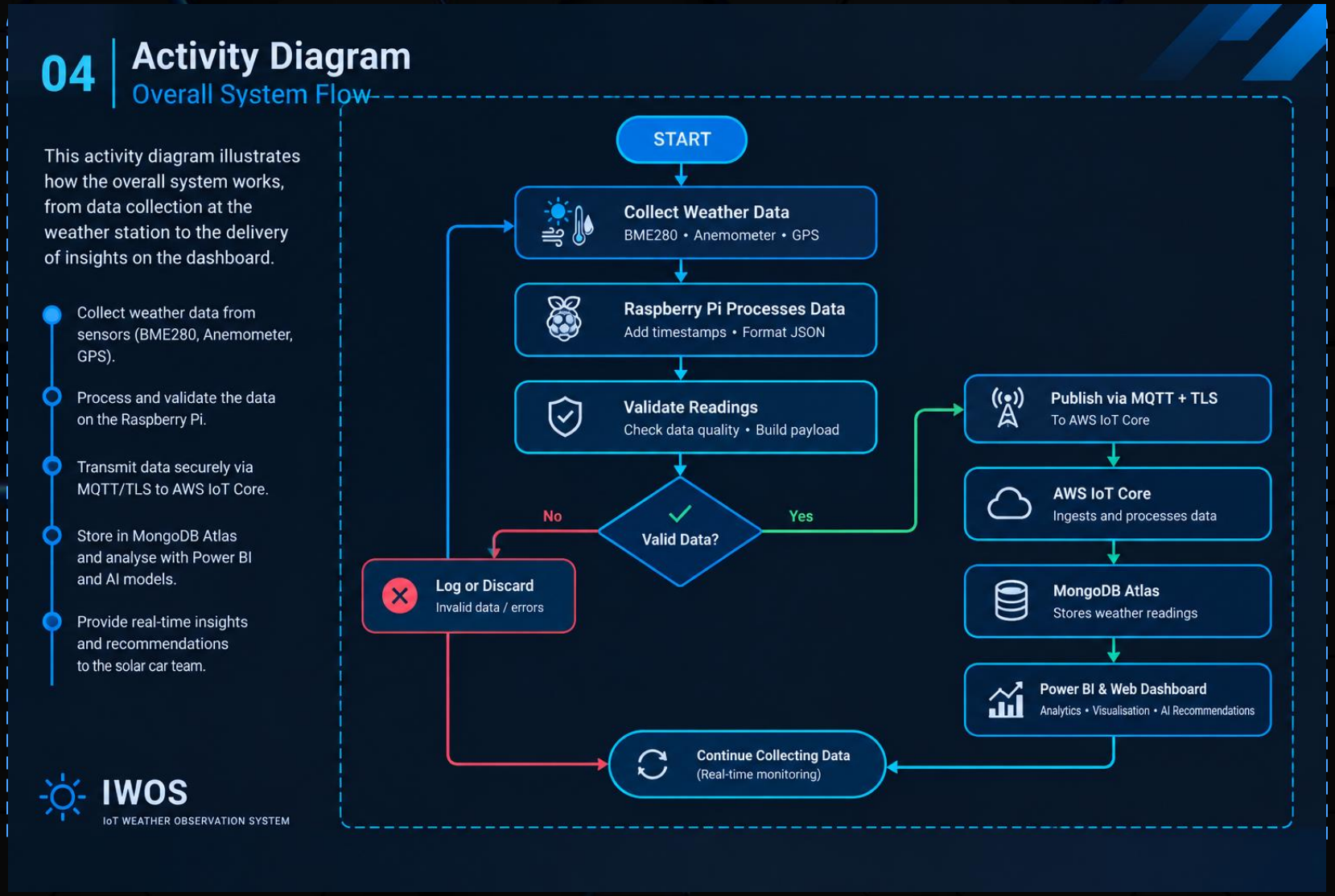
Overall system and subsystem



# 04 Activity diagram of overall system flow

## UML activity diagram

of how the overall system works



# 05 Code Snippets

Code snippet or screenshot goes here

# 06 Lessons Learnt

## Areas of Improvement

- Group Coordination
- Knowledge of tools used.
- Time Allocation/Planning

## Limitations

- Number of sensors (We can always get more data).
- Single server dependency
- Reliant on single service(AWS).
- Configuring secure MQTT communication with AWS IoT Core.
- Dependence on internet connectivity for real-time data transmission.

## Future Considerations

- Potential New Features.
- Multi-Platform Compatibility
- More Sensors = More Data = Better Optimization
- Test under-real world driving conditions

# 07 Overall Conclusion

## Project impact

- Helps to pre-plan routes.
- Optimize routes
- Collect real-world valuable data

## Who it benefits

- BC Sol Invictus Racing Team
- Belgium Campus

## New technologies learnt as a group

- Python(Sensor Array)
- MQTT & TLS (Data Transfer)
- AWS IoT Core (Connecting sensors to AWS cloud)
- Python Script ( PowerBi ), Figma(UI Design)

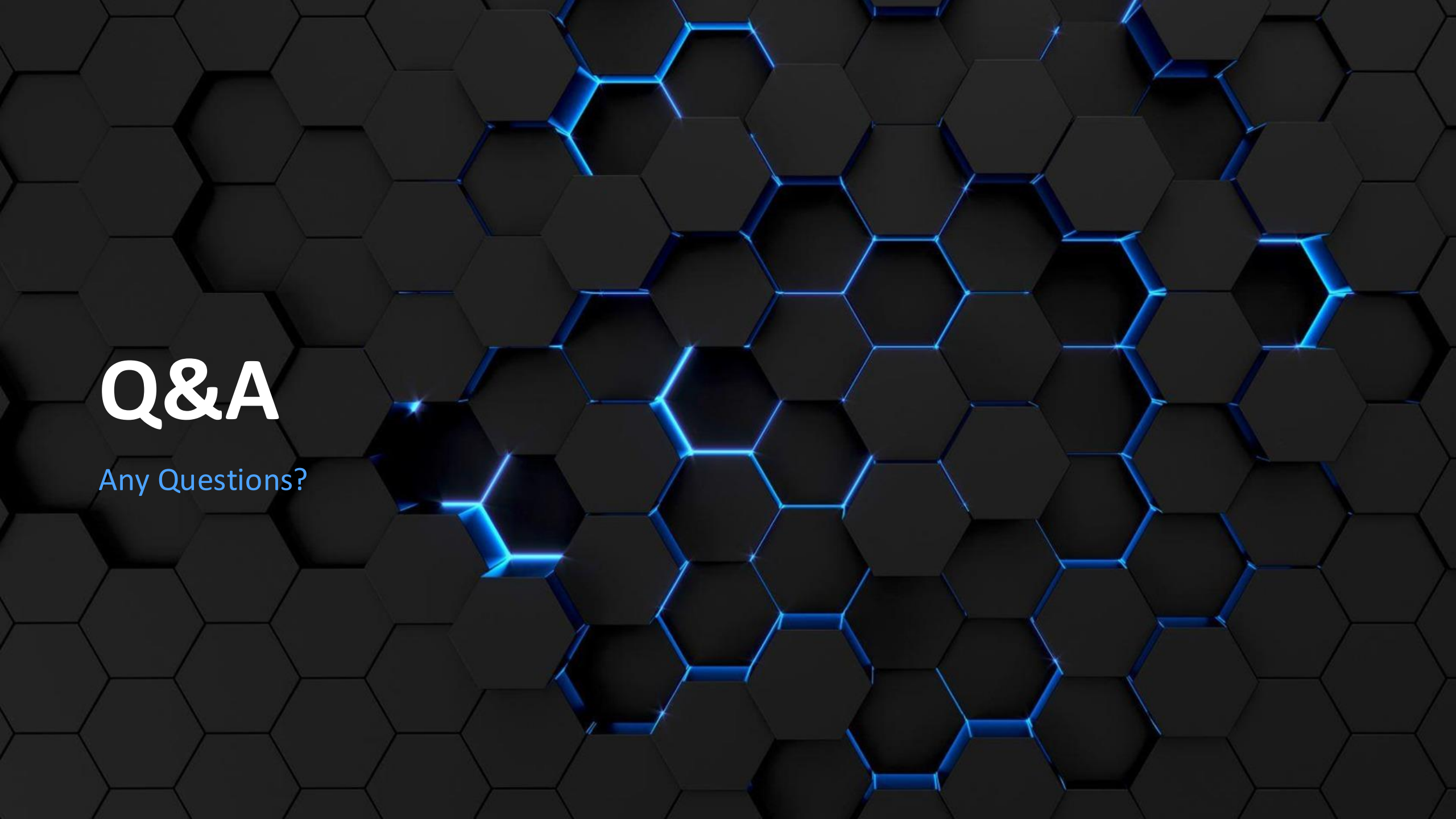
## Group dynamics and work ethics

- Diverse Group
- Different Areas Of Knowledge

# 08 Special Thanks or Mentions

## List those who assisted you in making the project a success

- Mr. Francios Venter – Helped Regardt to figure out how the GPS sensor worked.
- Claude AI- Helped Regardt find out what sensors we had.
- Claude AI- Helped me to connect the PowerBI to the MongoDB using Python Script.
- Special thanks to all Group 5 members — **Masana Mathonsi, Regardt Osler, Tebatso Letsoalo, Nkosana Mtsweni, Denzel Muranganwa, Lungisani Hadebe, Ulrich Rossouw, Phenyo Phooko, and Myself** (Sandile Machabane) — for their teamwork, dedication, and contributions towards making this project a success.


Q&A
Any Questions?